Device: Pulse Oximeter

Image from Google

1. What It Is

A pulse oximeter is a small, non-invasive, clip-like device typically attached to a thin part of the body—most often a fingertip, toe, or earlobe. It is used to rapidly measure two critical physiological parameters: oxygen saturation (SpO_2) and pulse rate.

2. Primary Function

Its core function is to monitor how efficiently the heart and lungs are delivering oxygen to the extremities. It answers two questions:

• **Oxygen Saturation (SpO_2):** What percentage of hemoglobin in the arterial blood is carrying oxygen? (Normal range: 95–100%)
• **Pulse Rate:** How fast is the heart beating? (Normal range: 60–100 bpm at rest)

By providing this data in real time, the device helps identify hypoxemia (low blood oxygen)—a dangerous, often silent condition that can precede organ failure or respiratory arrest.

3. How It Works (Physics in Practice)

The pulse oximeter uses two key principles: spectrophotometry and plethysmography.

• **Light Emission:** Two small LEDs inside the device emit light at specific wavelengths—red (660 nm) and infrared (940 nm).
• **Absorption Difference:** Oxygen-rich hemoglobin (oxyhemoglobin) absorbs more infrared light and allows more red light to pass through. Oxygen-poor hemoglobin (deoxyhemoglobin) does the opposite: it absorbs more red light and allows more infrared to pass.
• **Pulsatile Signal:** A photodetector on the other side of the finger measures how much light passes through. It isolates the signal from pulsatile arterial blood (the volume change with each heartbeat) from non-pulsatile tissue (skin, bone, veins).
• **Calculation:** A microprocessor compares the ratio of absorbed red to infrared light during the pulse, then uses a pre-calibrated algorithm to display the SpO_2 percentage.

4. Common Clinical Applications

· **Surgery & Anesthesia:** Continuous monitoring during and after operations.
· **Emergency Medicine:** Rapid triage of patients with trauma, shortness of breath, or chest pain.
· **Respiratory Diseases:** Monitoring COPD, asthma, pneumonia, or COVID-19 patients for silent hypoxia.
· **Sleep Studies:** Detecting drops in oxygen due to sleep apnea.
· **Home Use:** For patients on home oxygen or recovering from respiratory illness.

5. Key Limitations

• **Not a substitute for blood gas analysis:** It measures saturation, not partial pressure of oxygen (PaO_2) or carbon dioxide ($PaCO_2$).
• **False readings possible with:**
  · Dark fingernail polish or artificial nails.
  · Poor peripheral perfusion (cold hands, shock, vasoconstrictors).
  · Severe anemia (low hemoglobin gives misleadingly normal SpO_2).
  · Carbon monoxide poisoning (CO binds to hemoglobin and is read as oxygen).
• **Motion artifacts:** Patient movement can cause inaccurate readings.

6. Clinical Significance

The pulse oximeter is considered the fifth vital sign in many settings (after temperature, blood pressure, pulse, and respiration rate). Its immense value was highlighted during the COVID-19 pandemic, where it enabled early detection of hypoxemia before visible breathing difficulty. While not diagnostic, it provides a rapid, painless, and continuous “window” into cardiorespiratory status, saving countless lives through early intervention.

– Maimouna Tougoutcho Coulibaly